B · Before you begin

Read this chapter once before you pick up a drill. Budget a weekend: the steel on day one, the assembly on day two.

Safety

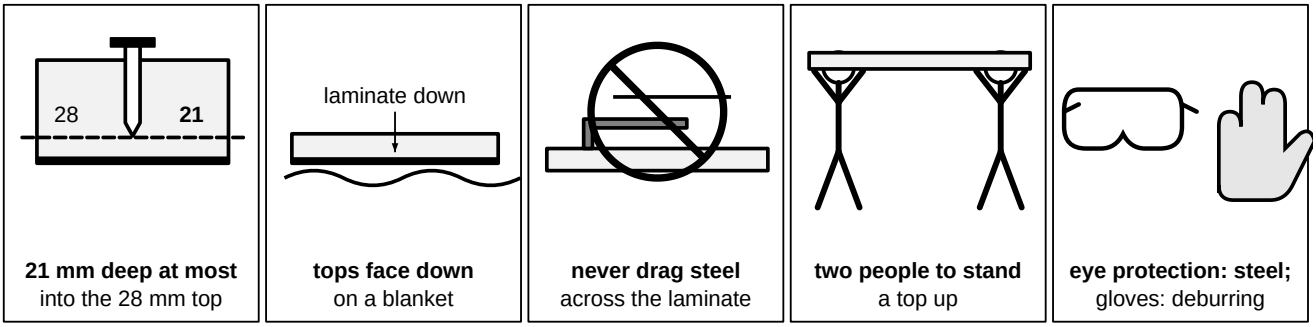
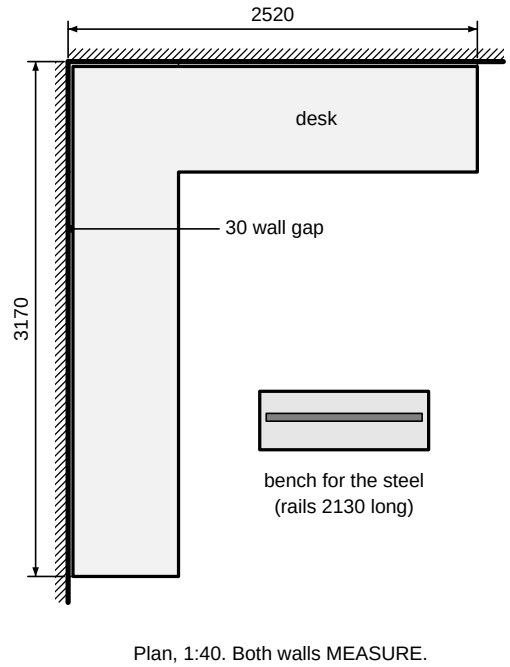


Fig. B-1 · The five rules that protect the tops and you.

> **NEVER A FASTENER DEEPER THAN 21 MM INTO A TOP**
> The EKBACKEN tops are 28 mm particleboard with a thin laminate skin. A screw or pilot that reaches the skin lifts it into a bump you will see and feel for ever. Every screw in this manual stops at least 7 mm short of the top face (28 − 21). If your tops measure 27 mm instead of 28, shorten every screw into the top by 1 mm. If you ever substitute a screw, keep this rule. Chapter D shows every depth.

- **Laminate protection.** Spread a blanket on the floor first. The tops lie face down on it, laminate on the blanket, for every marking and drilling step. Never slide or drag steel (rails, fittings, the clamp plates) across the laminate: lift it on and off.
- **Eye protection** for every hole drilled in steel; drill at low speed with cutting oil. **Gloves** while you deburr and file the steel: drilled edges are sharp.
- **Two people** to turn a top over and stand it on its legs (2490 long). A helper is also needed to slide the finished desk off and back to the walls.

Work area



Plan, 1:40. Both walls MEASURE.

Fig. B-2 · Floor needed: 2520 × 3170, including the 30 mm gaps to both walls.

The finished desk needs **2520 × 3170 mm** of clear floor in the corner: each top is 2490 long and stands 30 mm off its wall. MEASURE ON HARDWARE Measure both walls (wall to wall) and the baseboard thickness before you start.

You also need floor space for one top at a time lying face down on the blanket (2490 × 650 plus room to kneel all round), and a **bench** with two clamps for drilling the two 2130 mm steel rails. All steel is drilled on the bench, never under the desk.

Clear the corner before chapter G: the ALEX units are placed by the walls before the tops go on.

Contents

B B · Before you begin 1

C C · Complete parts inventory 6

D D · Component preparation 9

Workshop tools

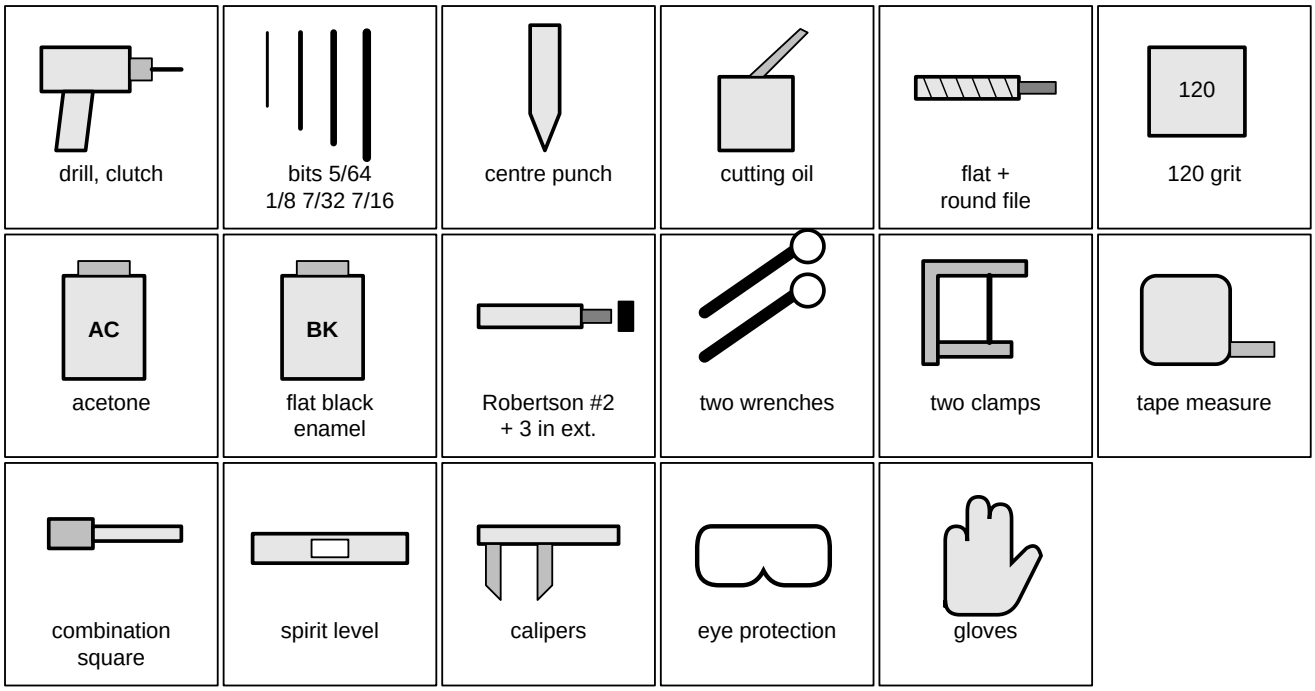


Fig. B-3 · Tools and protective equipment. Not to scale.

Tool	For
Corded or 18 V drill, clutch set low	all holes and screws
Bits: 5/64, 1/8, 7/32 in HSS; 7/16 in cobalt	ring and trim pilots; top and steel pilots; rail screw holes; fitting bolt holes
Centre punch (or a nail)	only for a mark the jig or plug cannot reach
Cutting oil or 3-in-1	every hole in steel
Flat file, round file, 120 grit, acetone, flat black enamel	deburr, ease a tight hole, scuff, degrease, paint the rails
Robertson #2 bit + 3 in magnetic extension	rail screws beside the vertical leg (the chuck would hit the leg otherwise)
Two wrenches or sockets for the 3/8-16 bolts (one for the head, one for the nut)	the corner fittings
Two clamps, 6 in or larger	hold each rail on its line; hold a fitting as a drilling template
Tape measure, combination square, masking tape, pencil	layout and the tape labels on the tops and rails
Spirit level, 60 cm or longer	setting the legs and feet
Calipers	top thickness, pad diameter, ALEX panel (the measurements on the last page of this chapter)

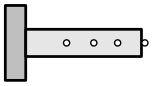
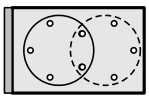
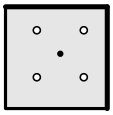
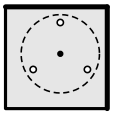
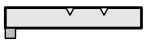
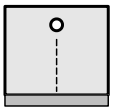
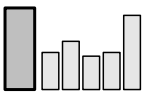
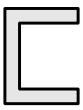
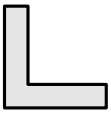
Protective equipment

- Safety glasses or goggles: every hole in steel, and opening the 7/16 in holes.
- Work gloves: deburring, filing and handling the cut rail ends.
- A blanket large enough for a whole top (2490 × 650).

Printed tools (P18): print these before you mark anything

All of them are in `parts/tools/`; `parts/README.md` has print settings. Print as saved, no supports. **Never scale a printed tool in the slicer: its distances are the tool.** The lipped and fenced tools print lip-up and are turned over to use.

> **EVERY JIG HOLE IS BIT-SIZED**
> Hold the jig on, start each pilot 3–4 mm through the jig's hole, lift the jig, then finish the hole to depth with the drill guide's puck. The small centre holes are for a scriber only; nothing is drilled through them.

Tool	What it does	Qty
	**Edge T-square.** Lip over any edge; punch holes at 50 (ring centres on the ALEX), 85 (feet, with the plate's four 1/8 in holes 46.8 apart round it), 100 (legs L2–L6) and 150 (leg L1).	1
	**Leg jig.** Lip over the wall edge; an OLOV's centre and its five 1/8 in holes in one placement: 100 pattern for L2–L6 (solid circle), 150 pattern for L1 (dashed circle, ringed holes).	1
	**Feet corner jig.** Fences on a far corner of a face-down top; a foot's centre 85/85 from both edges and its four 1/8 in holes 46.8 apart: F2, F4, F7, F8.	1
	**ALEX corner template.** Fences on an ALEX corner; ring centre 50/50, the ring's three 5/64 in holes (radius 33) and the three optional ALEX-foot holes (radius 14.5). Top and bottom corners.	1
	**Rail marking gauge.** Lip over the angle's edge; pencil notches at 19 (screw line on the flat leg) and 29 (bolt line on the vertical leg).	1
	**Rail drill jig.** Lip over the flat leg's free edge; a 1/8 in bore on the 19 line; sight grooves on the station line. PETG.	1
	**Slot plug** (fitting-centring-plug). Drops into a fitting's 21 × 12 slot; a 1/8 in bore dead centre, so the pilot cannot wander in the slot's play.	2
	**Pilot drill guide.** SET tower (45 for the 1/8 in bit, 35 for the 5/64 in) and five depth pucks: 12 PLATE, 15 LEG, 10 EDGE, 12 ALEX, 25 RAIL. Re-set the bit every time you change it.	1 set
	**Foot height gauge.** A C with 102 inside: all eight feet wound to the same height without a ruler.	1
	**Wall gauge.** An L: 65 (ALEX back to wall) and 30 (a top's raw edge to wall).	1

Also print the eight locating rings (P10) now: `parts/structure/foot-ring`, PETG or ABS, flange down. Their 51 mm bore depends on the pad diameter you measure (last page of this chapter).

How sure is each number?

Every dimension in this manual carries one of three confidence labels, or a fourth label that is an instruction to you:

VERIFIED	Measured on the purchased part, or taken from the part's own geometry file (the printed parts).
SOURCE-DEFINED	Defined by the project documents (the build manual): a design position, not a measurement. It is right by definition; you put the part there.
ASSUMPTION / FIELD CHECK	An engineering estimate or a check done on site. Confirm it before you rely on it.
MEASURE ON HARDWARE	Not yet measured on your parts. Measure it yourself before the step that uses it, and stop if it differs.

Confirm before you drill anything

> **MEASURE BEFORE DRILLING**
> These numbers are not yet confirmed on the parts you have. Each one is checked again where it is used; if one differs, stop and correct the layout before any hole is drilled.

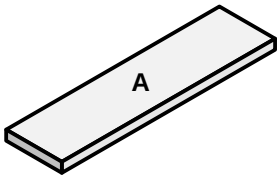
Measure	Expected	Label	Used in
EKBACKEN thickness, both tops (calipers)	28	MEASURE ON HARDWARE	D: every screw depth into the tops. At 27, shorten every screw into the top by 1 mm.
Delivered rail lengths, both pieces, and the angle thickness	2130 layout length (bought as 84 in = 2133.6); 4.8 thick	MEASURE ON HARDWARE	E: hole stations. 2130 is the layout length; the far end may carry the extra.
Strut fitting slots, positions from the heel	21 × 12 slots, centres 29.15 and 78.45 (legs 100.5 × 41)	VERIFIED	E, F: measured 23 Oct 2026; the fitting with the slot plug is still the drilling template.
Feet pad diameter (calipers)	50	MEASURE ON HARDWARE	G: the rings are bored 51 for a 50 pad; re-print them if yours differs.
ALEX top panel thickness, at an edge	16 or more for the 5/8 in screws	MEASURE ON HARDWARE	G: ring screws, 13 mm into the ALEX top.
ALEX height, with and without its stock feet	700	ASSUMPTION / FIELD CHECK	G: 700 + foot 102 = underside 802.
VIVO top arm reach; maximum clamp opening	about 89 ASSUMPTION / FIELD CHECK ; 31 or more	MEASURE ON HARDWARE	I: the clamps on the 28 top plus a pad plate.
Room: wall to wall along both walls; baseboard thickness	2520 and 3170	MEASURE ON HARDWARE	B (work area), G (placing the ALEX units by the walls).
Feet load rating	unknown; the design wants 50 kg or more each	ASSUMPTION / FIELD CHECK	C (P09); check the listing or the maker.

Already measured and printed as VERIFIED: the OLOV plate (105, five 5.5 holes on an 85 circle), the feet plate (59.5, four 5.2 holes 46.8 apart), the strut fittings (slots 21 × 12 at 29.15 and 78.45 from the heel), the VIVO crossbar (502 centre to centre), pole (35), pole inset (43.7), clamp back (80 × 17.5) and lower pad.

C · Complete parts inventory

Check every part against this list before you start. Part numbers P01–P21 and hardware numbers H01–H11 are used in every callout balloon in this manual. Dimensions in mm. REQUIRED parts are needed for the desk to stand; OPTIONAL parts can be left out.

thickness drawn ×4



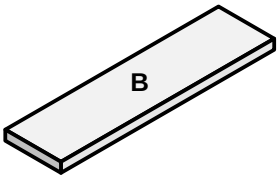
P01 **× 1**

Top A, IKEA EKBACKEN brown walnut, 2490 × 650 × 28. Front long edge rounded and laminated; back long edge square and unfinished (wall side).

MEASURE ON HARDWARE thickness

REQUIRED

thickness drawn ×4

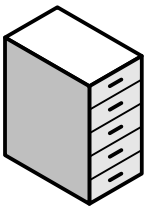


P02 **× 1**

Top B, identical to P01: 2490 × 650 × 28. Becomes B by being laid along the side wall (chapter D).

MEASURE ON HARDWARE thickness

REQUIRED



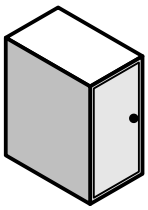
front

P03 **× 1**

ALEX 1, IKEA drawer unit, 360 × 580 × 700. Under top A's far end, drawers toward you.

ASSUMPTION / FIELD CHECK height 700

REQUIRED



front

P04 **× 1**

ALEX 2, IKEA storage unit with door, 360 × 580 × 700. Under top B's far end, door toward you.

ASSUMPTION / FIELD CHECK height 700

REQUIRED

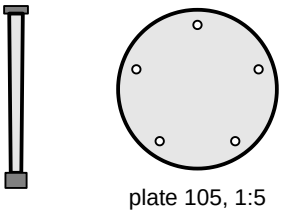


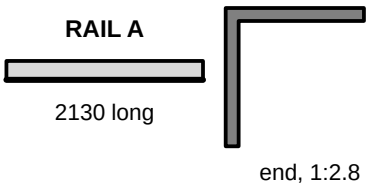
plate 105, 1:5

P05 **× 6**

IKEA OLOV leg, adjustable 600–900, set to 802. Plate ⌀105, five ⌀5.5 holes on an ⌀85 circle. Max 13 kg each.

VERIFIED plate

REQUIRED

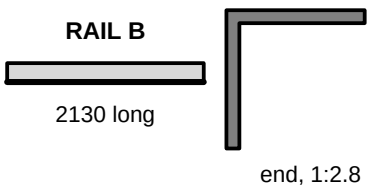


P06 **× 1**

Rail A, 2 × 2 × 3/16 in hot-rolled steel angle (50.8 × 50.8 × 4.76), 2130 layout length (bought as 84 in = 2133.6).

MEASURE ON HARDWARE length

REQUIRED

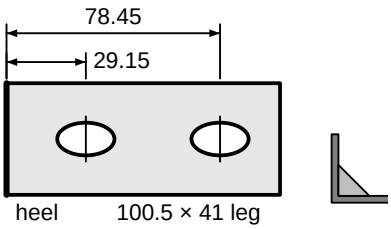


P07 **× 1**

Rail B, identical to P06 until its corner end is taped (chapter D).

MEASURE ON HARDWARE length

REQUIRED

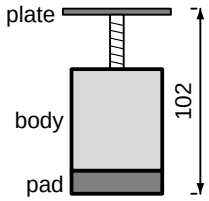


P08 **× 2**

Gusseted 90° strut fitting, 5 mm steel, for 1-5/8 in channel. Legs 100.5 × 41; two 21 × 12 slots per leg, centres 29.15 and 78.45 from the heel.

VERIFIED

REQUIRED



P09 **× 8 (+4)**

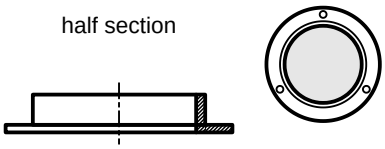
Round adjustable foot, 4 in, set to 102. Body and pad ⌀50; plate 59.5 square, four ⌀5.2 holes 46.8 apart. 12 on hand: 8 used, 4 SPARE.

VERIFIED plate ·

MEASURE ON HARDWARE pad · load

ASSUMPTION / FIELD CHECK

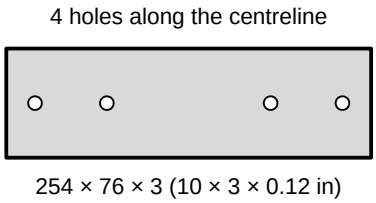
REQUIRED



P10 **× 8**

Locating ring, printed PETG or ABS: bore 51, OD 57, ring 10 high on a ⌀75 × 2.5 flange, three countersunk ⌀3.8 holes on radius 33. **VERIFIED** geometry file

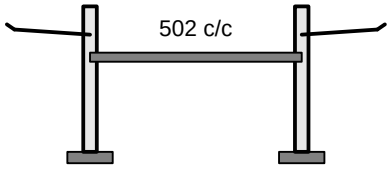
REQUIRED



P11 **× 4**

Mending plate, 10 × 3 × 0.12 in, holes at 20, 70, 184, 234 from one end. Two join the tops (P1, P2); two pad the VIVO clamp jaws. **SOURCE-DEFINED**

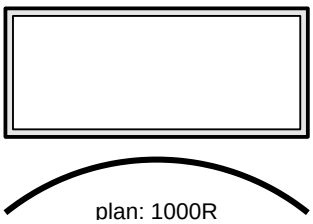
REQUIRED



P12 **× 1**

VIVO STAND-V300CU: two clamps, two ⌀35 poles 610 (24 in) tall, crossbar 502 centre to centre, two side arms, centre bracket. Clamp back 80 along the edge, 17.5 behind it. **VERIFIED**

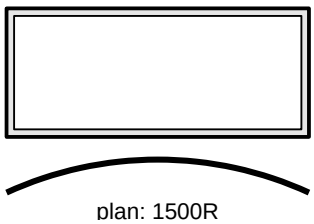
REQUIRED



P13 **× 1**

Samsung Odyssey G5 32 in, 1000R, VESA 75, about 5 kg. Centre screen, landscape, bottom edge 230 above the top. **SOURCE-DEFINED**

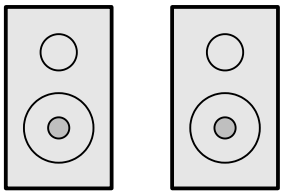
REQUIRED



P14 **× 2**

Acer XZ320QU 31.5 in, 1500R, VESA 100, 5.6 kg with stand (about 5 kg on the arm). Side screens, landscape. **SOURCE-DEFINED**

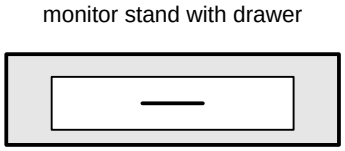
REQUIRED



P15 **× 2**

Edifier R1280DBs speakers, 234 tall, one under each Acer. **SOURCE-DEFINED**

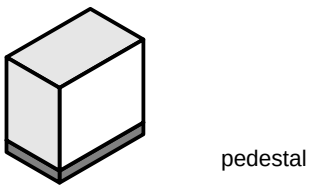
REQUIRED



P16 **× 1**

IKEA ELLOVEN monitor stand with drawer, 100 tall, under the G5. **SOURCE-DEFINED**

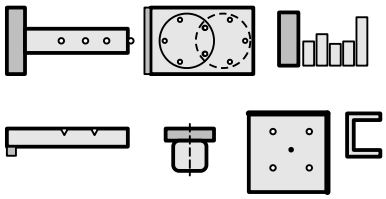
REQUIRED



P17 **× 1**

HYTE Y60 on its printed pedestal (MakerWorld 679012): footprint 490 × 310, 480 tall with the pedestal, over ALEX 1. **SOURCE-DEFINED**

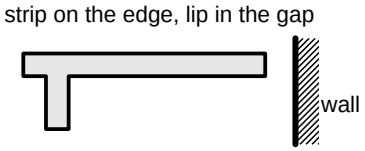
REQUIRED



P18 **1 set**

Printed tools: T-square, leg jig, feet corner jig, ALEX corner template, rail gauge, rail drill jig, slot plug × 2, drill guide (tower + 5 pucks), foot height gauge, wall gauge. List in chapter B.

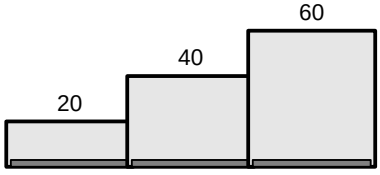
REQUIRED



P19 **1 set**

Wall-gap trim, printed: corner, fillers, clamp pieces, segments, connector tiles. Cut list in `parts/README.md`. Fixed with H07.

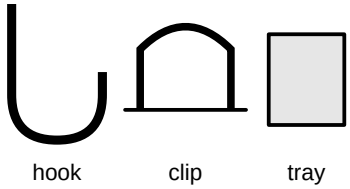
OPTIONAL



P20 **× 8**

ALEX feet, printed: ⌀56, one height for all (20, 40 or 60), with TPU pads. Raise the whole desk by their height.

OPTIONAL



P21 **1 set**

Cable parts, printed: rail cable hooks (magnets H11), cable clips, power-bar cradles, cable trays.

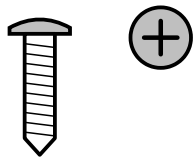
OPTIONAL

Drawings are silhouettes for recognition, not to scale unless a scale is written on them.

Hardware H01–H11

Screws and bolts are drawn **1:1** (true length, nominal diameter): lay a screw on the drawing to check it. Washers and the nut are symbols. All hardware was ordered 22 Oct 2026 SOURCE-DEFINED. Buy a few spares of each screw.

H01 × 30

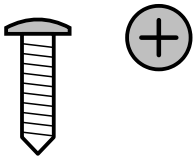


1:1 · head as supplied

M4 × 16 machine screw, fully threaded, IKEA 108443 (with the legs). OLOV plates, 5 per leg.

REQUIRED

H02 × 32

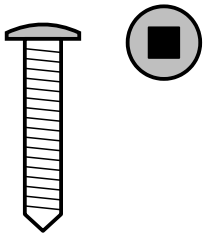


1:1 · pan head

#8 × 5/8 in pan head. Feet plates, 4 each.

REQUIRED

H03 × 19

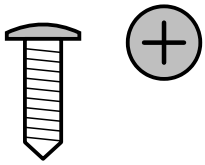


1:1 · pan head, Robertson

#10 × 1 in pan head, Robertson. Rails into the tops (9 + 10).

REQUIRED

H04 × 8

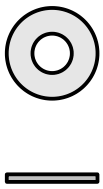


1:1 · pan head

#10 × 5/8 in pan head. Joining plates P1, P2, 4 each.

REQUIRED

H05 × 8

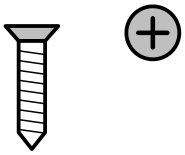


symbol, not to scale

#10 flat washer, under each H04.

REQUIRED

H06 × 24

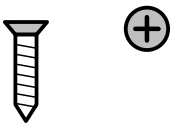


1:1 · flat head

#6 × 5/8 in flat head. Rings, 3 each. +24 for ALEX feet, more for cable parts.

REQUIRED

H07 × 48

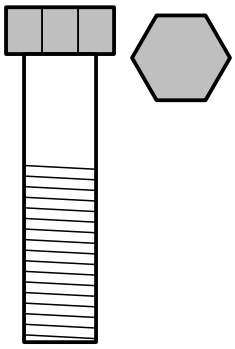


1:1 · flat head

#4 × 1/2 in flat head. Wall-gap trim, two per piece.

OPTIONAL

H08 × 6

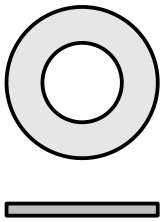


1:1 · hex head

3/8-16 × 1-1/2 in hex bolt, grade 5. Corner fittings.

REQUIRED

H09 × 12

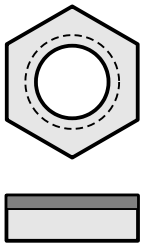


symbol, not to scale

3/8 flat washer, under each bolt head and each nut.

REQUIRED

H10 × 6

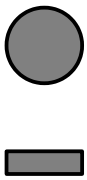


symbol, not to scale

3/8-16 nylon-insert lock nut. Stiff to turn; that is normal.

REQUIRED

H11 × 8



1:1

10 × 3 mm neodymium disc magnet, two per rail cable hook.

OPTIONAL

D · Component preparation

This chapter marks both tops face down and labels the two rails. Nothing is fixed yet. Every position is in mm from one **datum**: **top A's back-left corner**. **x** runs along the back wall, **y** along the side wall. Top A covers x 0–2490, y 0–650; top B covers x 0–650, y 650–3140.

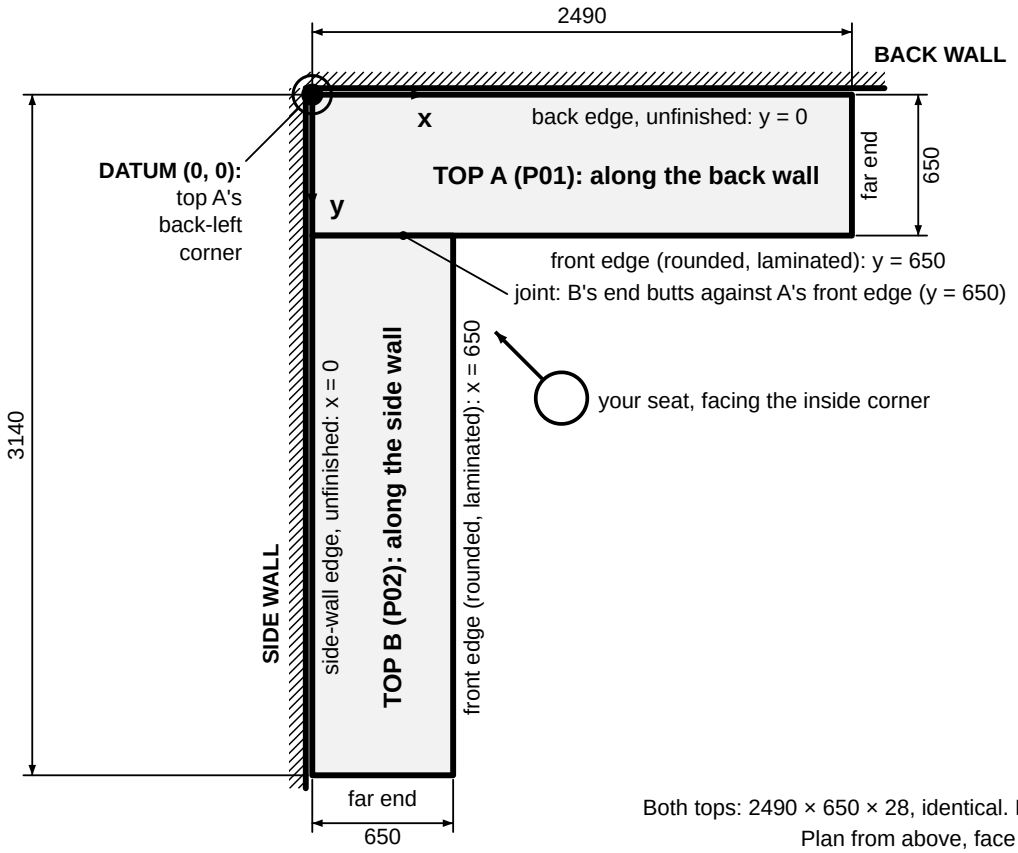


Fig. D-1 · The datum and the two tops, seen from above as installed. SOURCE-DEFINED

Which edge is which

Each top has one **rounded, laminated** long edge: the **front**. The other long edge is **square and unfinished**: it goes against the wall. Either short end can become the side-wall end.

Which top is A

The two tops (P01, P02) are identical. **A** is the one you lay along the back wall; its unfinished edge is y = 0. **B** lies along the side wall; its unfinished edge is x = 0 and one short end, the **joint end** (y = 650), butts against A's front edge. Top B is top A turned 90°, never mirrored.

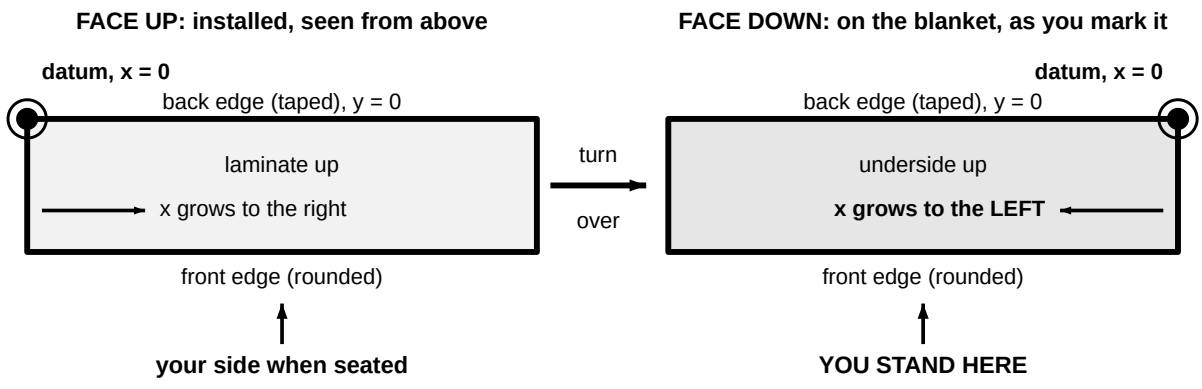
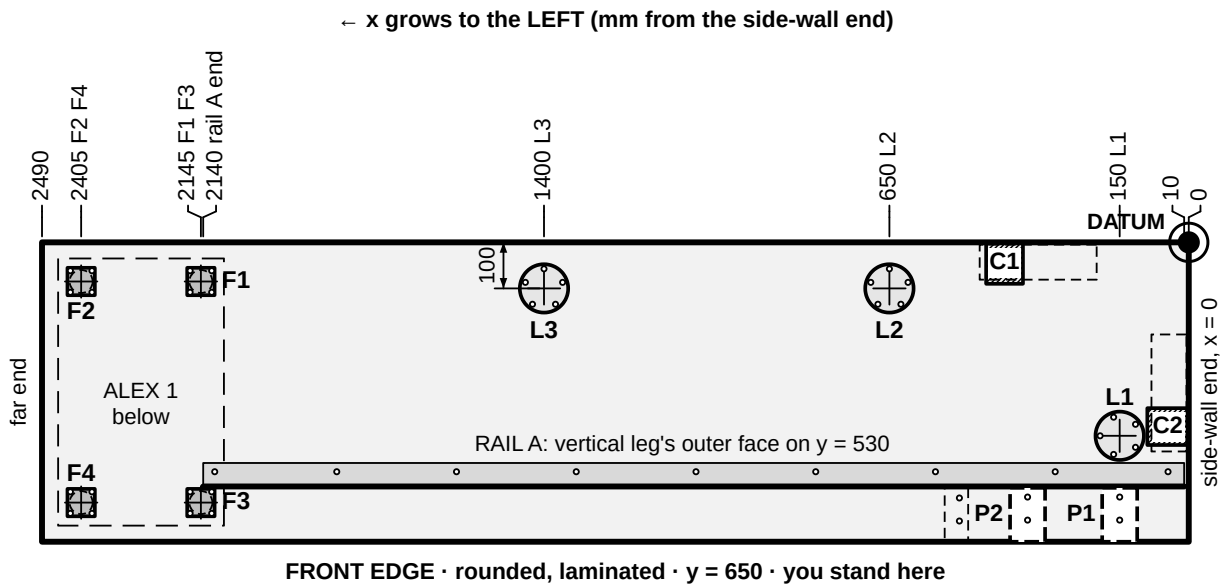


Fig. D-2 · Face-down convention used on every marking drawing. Stand opposite the taped reference edge (top A: at its front edge; top B: at its far end, joint end away from you): the underside is the plan mirrored left to right, so **x grows to the LEFT and the datum side (x = 0) is on your RIGHT**. Top B uses the same mapping.

Top A, face down: overview

1 Calipers on top A: 28 mm MEASURE ON HARDWARE. If it is 27, shorten every screw into it by 1 mm. Tape the **unfinished long edge** ("A · BACK EDGE") and the short end you choose as the **side-wall end** ("A · SIDE-WALL END"). Lay top A face down on the blanket and stand at the front edge, back edge away from you.



Scale 1:16. Corner and far end in Figs D-4 and D-5.

Fig. D-3 · Top A underside as you see it. Datum at the far right corner. Circles: OLOV plates (P05) with five pilots; squares: feet plates (P09) with four pilots; grey band: rail A (P06) with its screw positions; hatched C1, C2: clamp zones; dashed: pad plates, joining plates P1, P2 (P11), rail B's corner end and ALEX 1 (reference).

Mark	Centre (x, y)	Tool and placement	Note
L1	(150, 420)	leg jig, **150** pattern, lip over the **side-wall end**, station y = 420	under VIVO clamp 2; plate ends 6.5 short of rail A
L2	(650, 100)	leg jig, 100 pattern, lip over the back edge, station x = 650	nearest leg to clamp 1
L3	(1400, 100)	leg jig, 100 pattern, lip over the back edge, station x = 1400	
F1	(2145, 85)	T-square hole 85 and its four plate holes, lip over the back edge, station x = 2145	feet for ALEX 1, 50 in from each ALEX corner; plates 5 clear of rail A
F2	(2405, 85)	feet corner jig on the far back corner	
F3	(2145, 565)	T-square, lip over the **FRONT** (rounded) edge, station x = 2145	
F4	(2405, 565)	feet corner jig on the far front corner	
Rail A line	y = 530, x 10–2140	pencil line; the vertical leg's outer face goes on it	flat leg covers y 479–530
C1 / **C2**	x 360–440, y 0–90 / x 0–90, y 360–440	keep clear: VIVO clamp zones	nothing fixed here
Pad plates	x 200–454, y 5–81 / x 5–81, y 200–454	outline only	loose P11 plates under the clamp jaws, chapter I
P1 / **P2**	x 112–188 / x 312–388, y 533–650	outline only (each plate spans y 533–787 across the joint)	3 clear of rail A; screws later at y 553 and 603

All positions SOURCE-DEFINED (docs/build-manual.md step 2). Pilots: 1/8 in, started 3–4 mm through the jig; finishing them to depth and fitting the legs and feet follows in the next chapter.

Top A: the corner

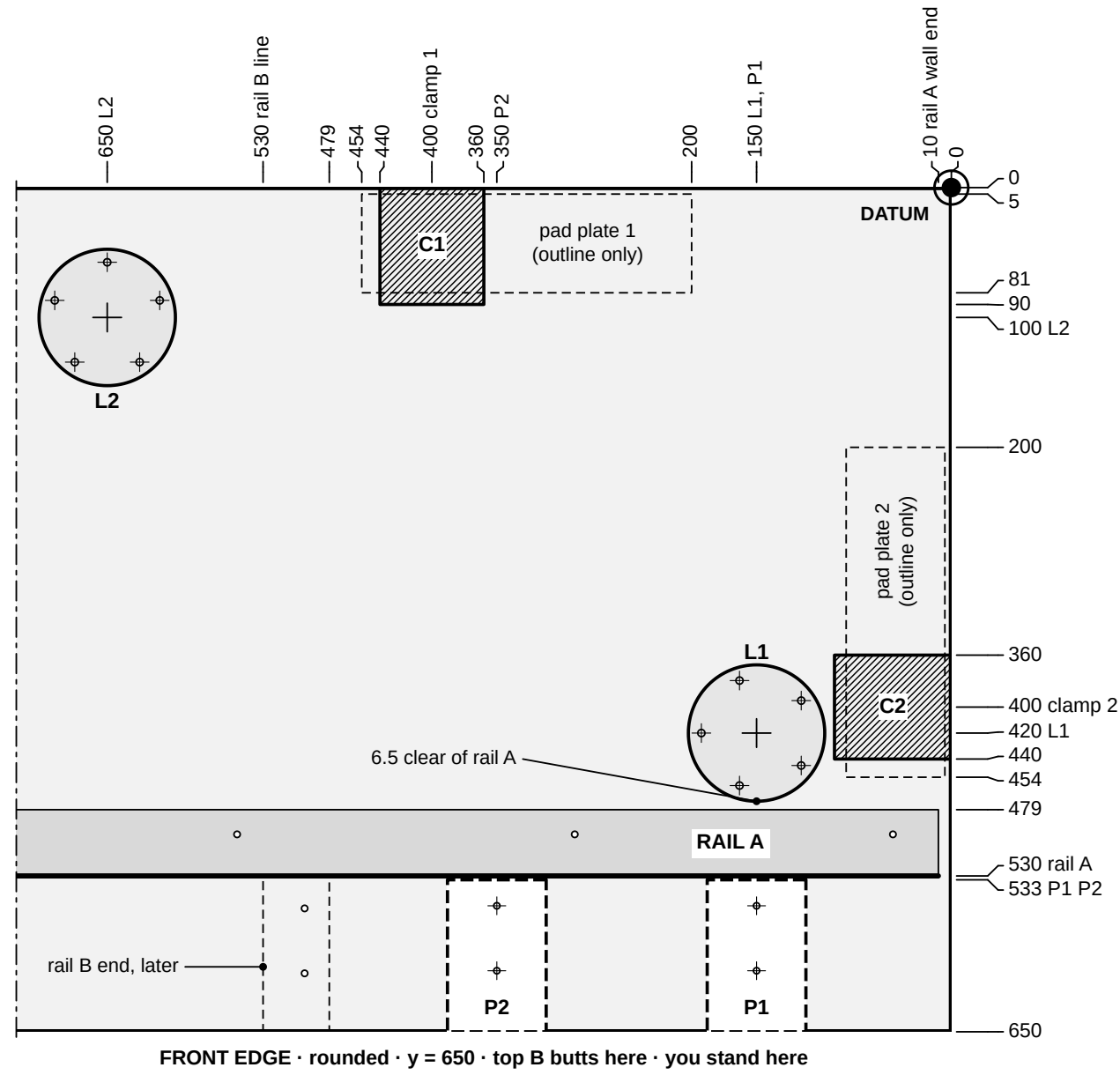


Fig. D-4 · Top A corner (x 0–720), face down, 1:5. Every station is measured from the datum: x along the back edge from the side-wall end, y in from the back edge.

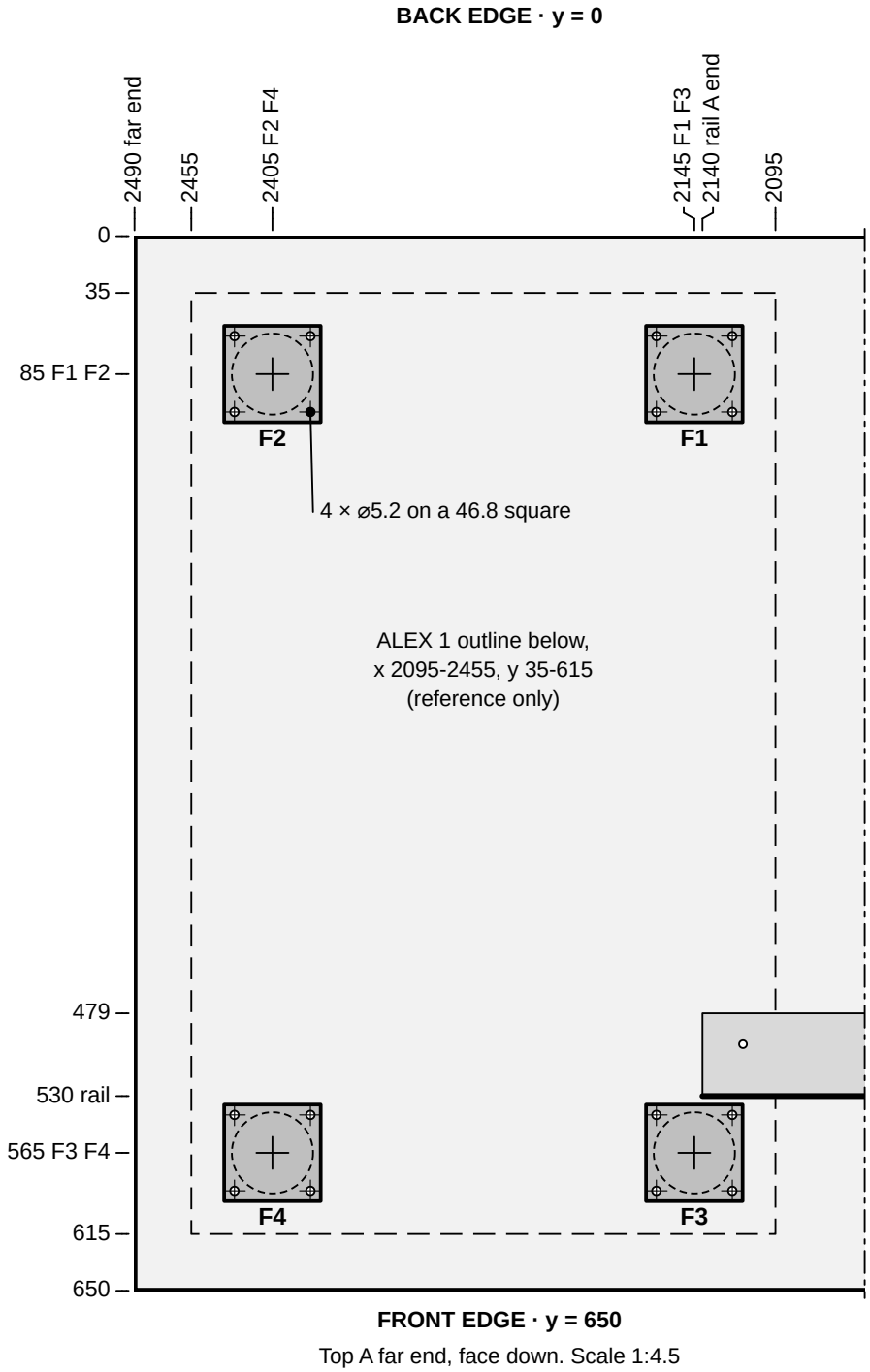
2

L1: hook the leg jig's lip over the **side-wall end** (x = 0), slide it to station **y = 420**, and use its **150** pattern (dashed circle, ringed holes): start the five 1/8 in pilots. **L2:** lip over the back edge at x = 650, **100** pattern (solid circle).

3

Draw the rail A line at **y = 530** from x = 10 to 2140 (both ends and the middle measured from the back edge). Hatch C1 and C2 with pencil. Draw the two pad-plate outlines and the P1, P2 outlines. Nothing is drilled inside C1, C2 or the outlines.

Top A: the far end (feet for ALEX 1)



Top A far end, face down. Scale 1:4.5

Fig. D-5 · Top A far end (x 2040–2490), face down, 1:4.5.

4 **F2, F4:** drop the feet corner jig over the far back corner, then the far front corner, fences against both edges. Start the four 1/8 in pilots through it.

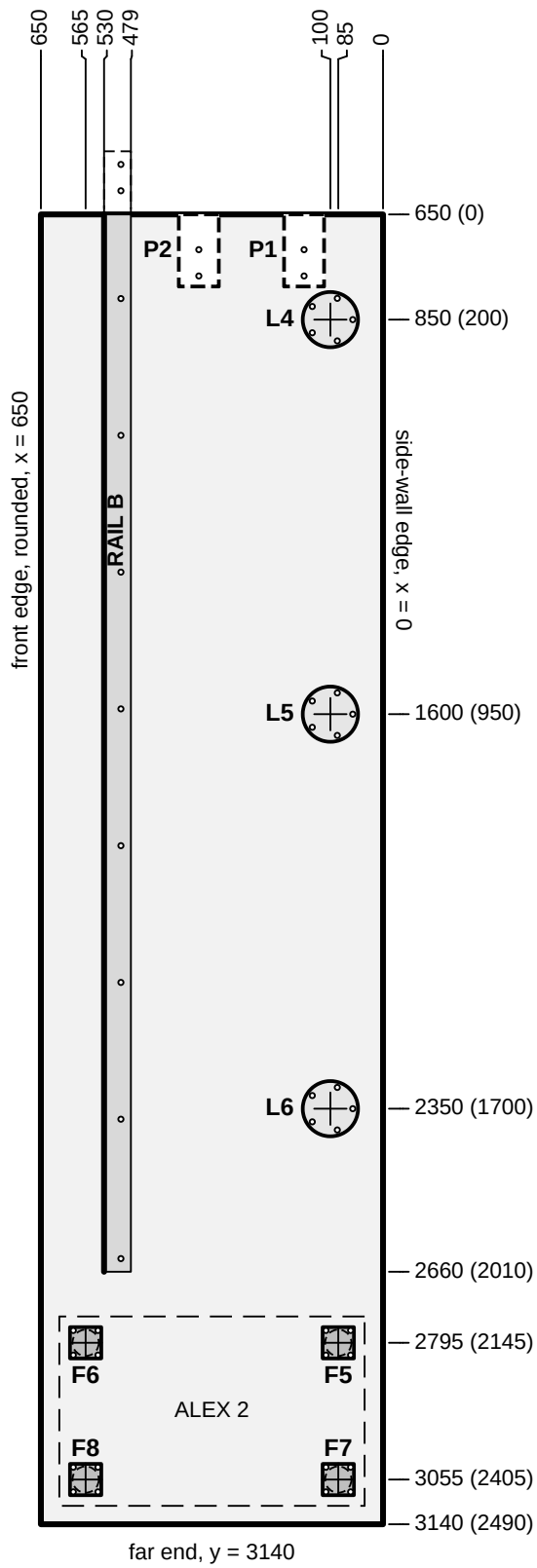
5 **F1:** T-square, lip over the back edge at station x = 2145: scriber through hole 85 to check, then start the four pilots round it. **F3:** the same with the lip over the **front** edge.

6 **L3:** leg jig, 100 pattern, lip over the back edge at x = 1400 (Fig. D-3).

Feet plate: 59.5 square, four ⌀5.2 holes 46.8 apart **VERIFIED**. Plates square with the top's edges.

The ALEX 1 outline (x 2095–2455, y 35–615) is for orientation only; nothing is marked for it.

Top B, face down: overview



y labels: desk y (tape from the joint end)
Scale 1:14

Fig. D-6 · Top B underside, joint end away from you, 1:14.

7

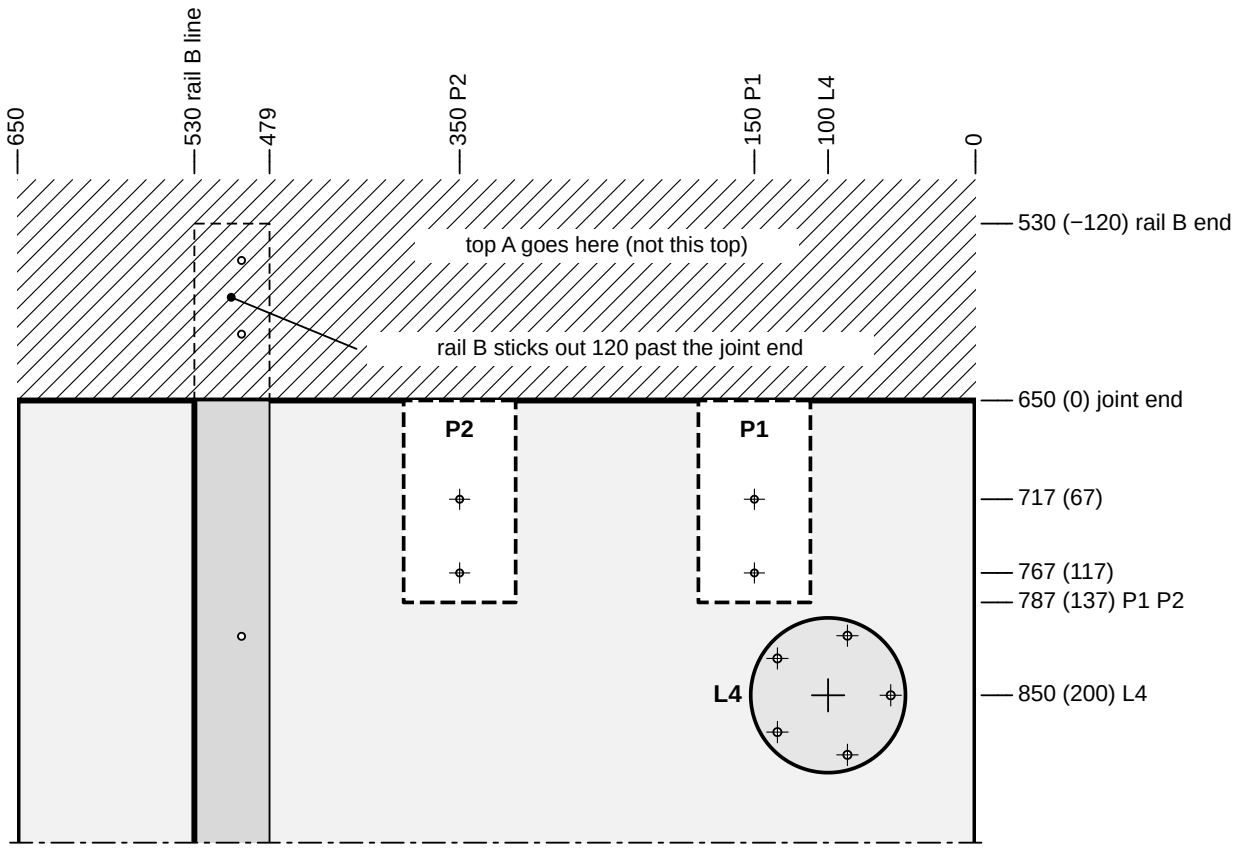
Calipers on top B: 28 mm MEASURE ON HARDWARE. Tape its **unfinished long edge** ("B · SIDE-WALL EDGE", x = 0) and the short end that will butt against top A ("B · JOINT END", y = 650). Lay it face down and stand at the far end, joint end away from you: the side-wall edge is on your right.

> Desk y on top B starts at 650. Hook the tape on the **joint end**: it reads **y − 650**, the number in brackets.

Mark	(x, y)	Tape from joint end	Tool and placement
L4	(100, 850)	200	leg jig, 100 pattern, lip over the side-wall edge
L5	(100, 1600)	950	
L6	(100, 2350)	1700	
F5	(85, 2795)	2145	T-square 85, lip over the side-wall edge
F6	(565, 2795)	2145	T-square 85, lip over the **FRONT** edge (x = 650)
F7	(85, 3055)	2405	feet corner jig, far wall-side corner
F8	(565, 3055)	2405	feet corner jig, far front corner
Rail B line	x = 530, y 650–2660	0–2010	vertical leg's outer face on it; flat leg x 479–530; corner end sticks out 120 past the joint end
P1 / **P2**	x 112–188 / 312–388, y 650–787	0–137	outline only; screws later at y 717 and 767 (67, 117)

All positions SOURCE-DEFINED (docs/build-manual.md step 3). L4–L6 have one pilot pointing at the side wall, like L2 and L3.

Top B: the joint end



Top B joint end, face down: joint end away from you, x grows to the left. Scale 1:5
(…) = tape reading from the joint end. P1, P2 holes: screw positions.

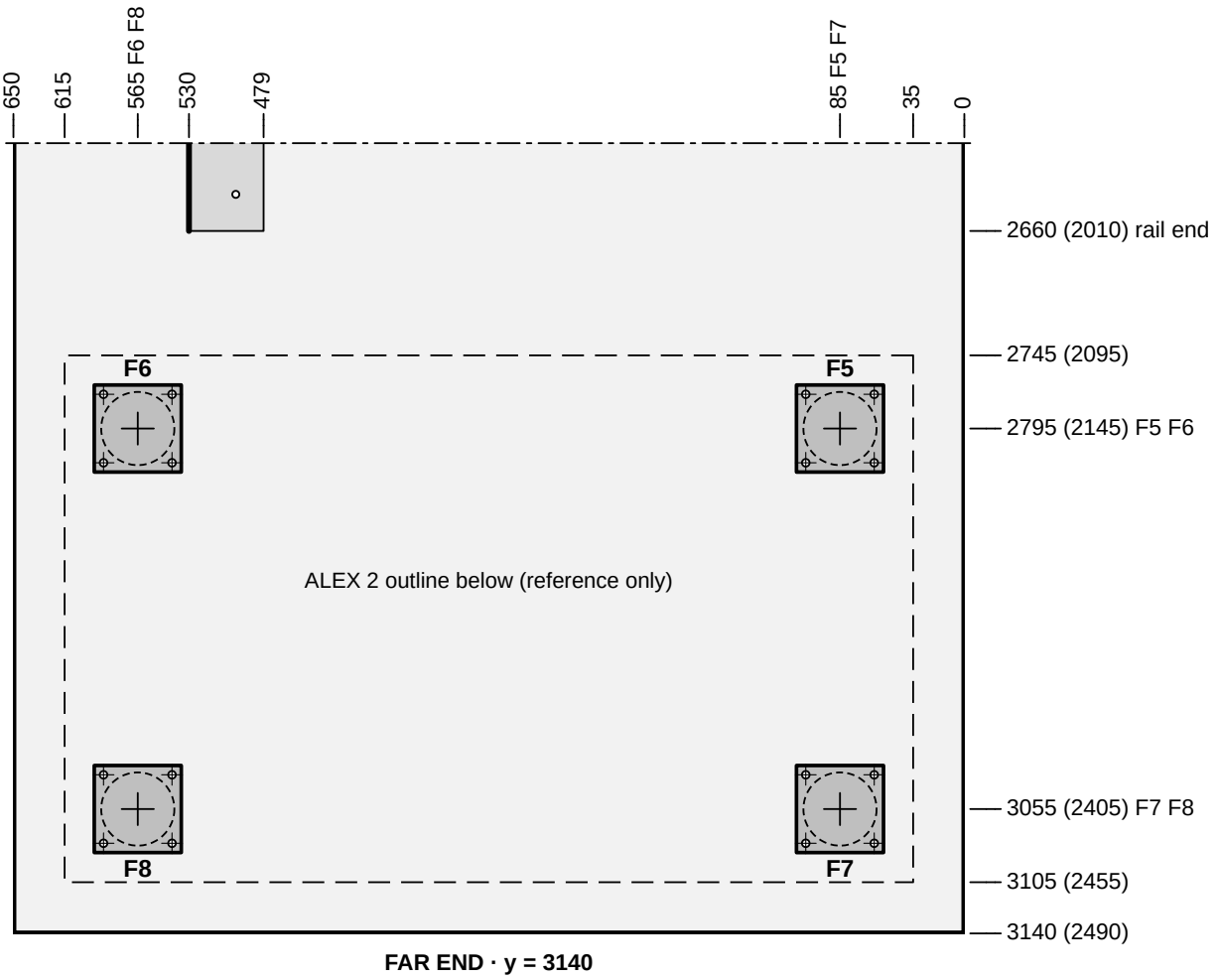
Fig. D-7 · Top B joint end (y 500–950), face down, 1:5. The hatched area beyond the joint end is where top A will be; rail B's corner end lies there.

8

L4: leg jig, 100 pattern, lip over the side-wall edge, tape 200 from the joint end. Draw the **rail B line at x = 530** from the joint end to tape 2010 (y 2660). Draw the P1 and P2 outlines from the joint end to tape 137 (y 787).

Rail B's holes 1 and 2 (shown beyond the joint end) go into top A when the tops are joined; on top B only its holes 3 to 10 are used.

Top B: the far end (feet for ALEX 2)



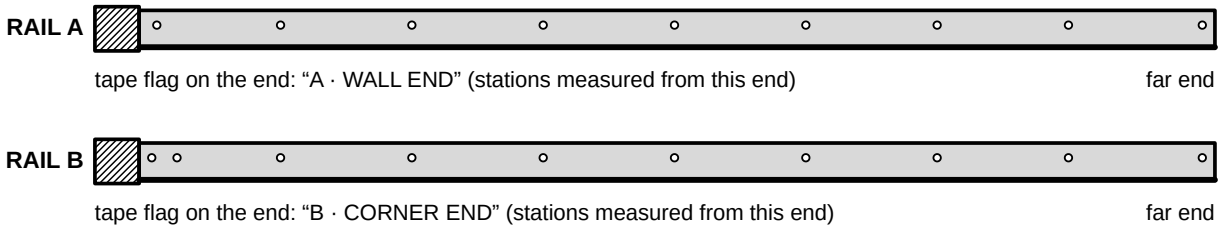
Top B far end, face down. Scale 1:5. (…) = tape from the joint end

Fig. D-8 · Top B far end (y 2600–3140), face down, 1:5.

9 **F7, F8:** feet corner jig over the two far corners, fences against both edges; start the four 1/8 in pilots. **F5:** T-square, lip over the side-wall edge, tape 2145 from the joint end; **F6:** the same with the lip over the front edge. **L5, L6:** leg jig, 100 pattern, tape 950 and 1700 (Fig. D-6).

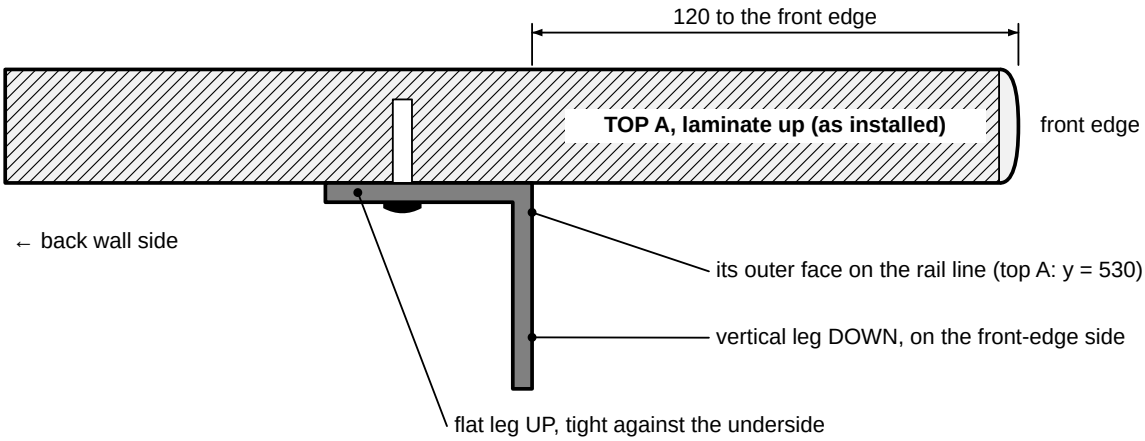
Rails: labels and orientation

10 The two steel angles arrive identical; either can be rail A. Tape one end of each: **"A · WALL END"** and **"B · CORNER END"**. Every station on a rail is measured from its taped end. Measure both delivered lengths MEASURE ON HARDWARE (2130 layout length; the far end may carry the extra).



Bench view, flat leg up. Holes shown for orientation; positions are in chapter E.

Fig. D-9 · The two rails on the bench, flat legs up, labelled ends left.



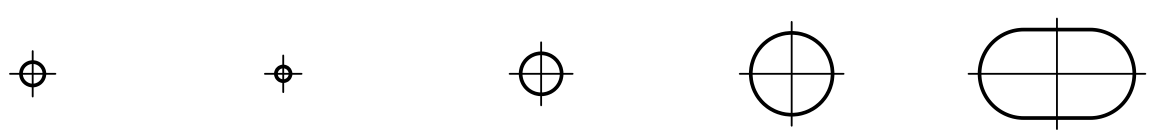
Section through top A and rail A, end-on, about 1:2. Rail B on top B: the same, with its outer face on x = 530.

Fig. D-10 · How a rail sits under its top: **flat leg UP** against the underside, all screws through it; **vertical leg DOWN** on the front-edge side, its outer face on the rail line, 120 from the front edge. Rail A: outer face on y = 530, wall end at x = 10. Rail B: outer face on x = 530, corner end 120 past top B's joint end.

The printed rail end caps are handed: cap A on rail A's far end and rail B's far end, cap B on rail A's wall end; rail B's corner end takes no cap.

Hole types and depths

The three hole types



pilot ⌀3.2 (1/8 in) into the tops: legs, feet, rails, joining plates

pilot ⌀2.0 (5/64 in) into the ALEX tops and the trim edge

clearance ⌀5.56 (7/32 in): rail flat legs, for #10 screws

through ⌀11.1 (7/16 in) rail vertical legs, 6 holes, 3/8 bolts

slot 21 × 12 in each fitting leg, bought, not drilled

Fig. D-11 · Pilot holes go into wood; clearance and through holes go through steel. The fittings come with 21 × 12 slots (not drilled); the slot plug centres the rail's 7/16 in pilot in them.

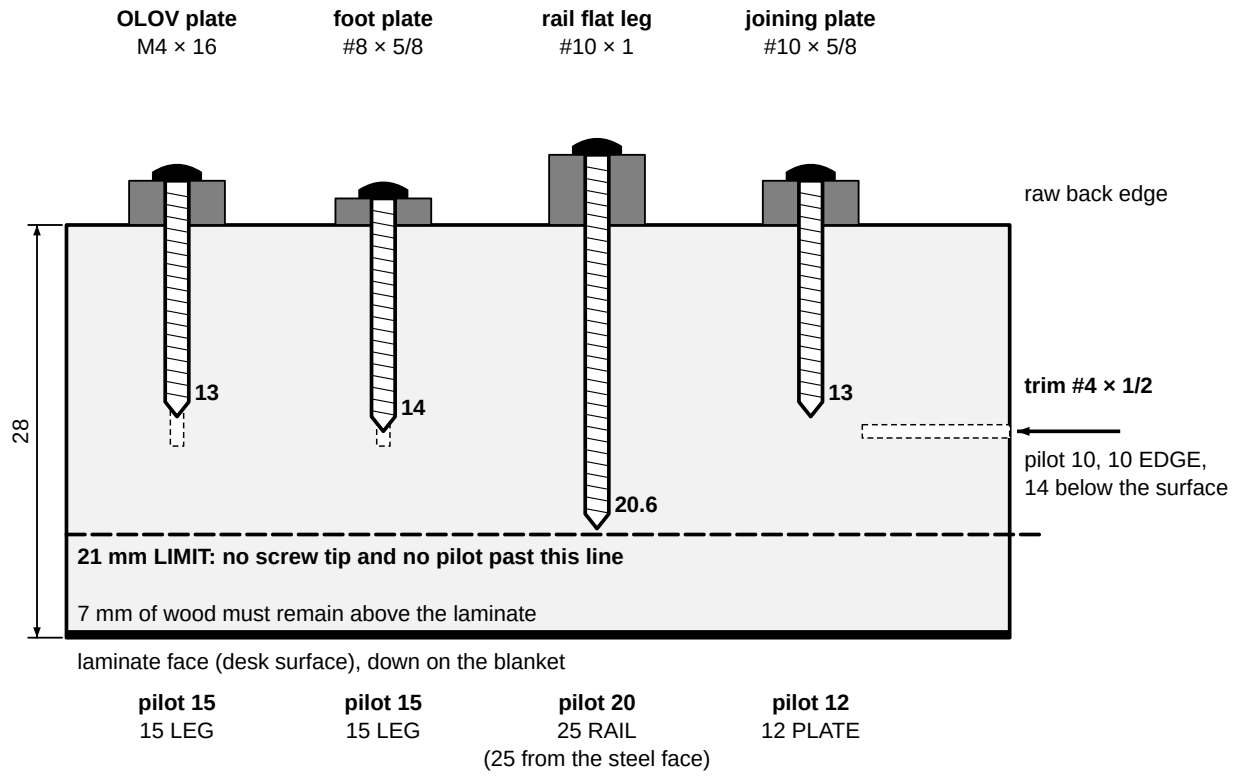
Every fastener, its pilot and its puck

Fastener	Qty	Goes through	Into the wood	Pilot	Puck
H01 · M4 × 16 (with the OLOV)	30	3 mm leg plate	13	1/8 in, 15 deep	15 LEG
H02 · #8 × 5/8 in pan head	32	feet plate (1.5–2 mm)	14	1/8 in, 15 deep	15 LEG
H03 · #10 × 1 in pan head, Robertson	19	4.8 mm angle	20.6	1/8 in, 25 from the flat leg's face = 20 in the wood	25 RAIL
H04 + H05 · #10 × 5/8 in with washer	8	3 mm joining plate	13	1/8 in, 12 deep	12 PLATE
H06 · #6 × 5/8 in flat head	24	2.5 mm ring flange	13 into the ALEX top	5/64 in, 12 deep	12 ALEX
H07 · #4 × 1/2 in flat head (optional trim)	48	trim lip	sideways into the raw back edge, 14 below the surface	5/64 in, 10 deep	10 EDGE
H08 + H09 + H10 · 3/8-16 bolts	6	fittings and rail vertical legs	none	7/16 in, bench-drilled	—

Depths SOURCE-DEFINED (guide.html fastener table, facts.json). The bolt holes and the 7/32 in screw holes in the steel are drilled on the bench (chapter E); everything into wood goes through a jig first and is finished with its puck.

Never through the laminate

NEVER THROUGH THE LAMINATE
No screw tip and no pilot goes deeper than **21 mm** into a 28 mm top: **28 − 21 = 7 mm** of wood must remain under the laminate face. Set every pilot depth with the drill guide's puck, never by eye. If a top measures 27 mm, shorten every screw into it by 1 mm.



Section through a 28 mm top, face down as you drill it (underside up), 2:1. Grey: the part being screwed on. Dashed: the pilot hole. Bold numbers: how far the screw goes into the wood, mm.

Fig. D-12 · Every fastener into the tops at its depth, 2:1.

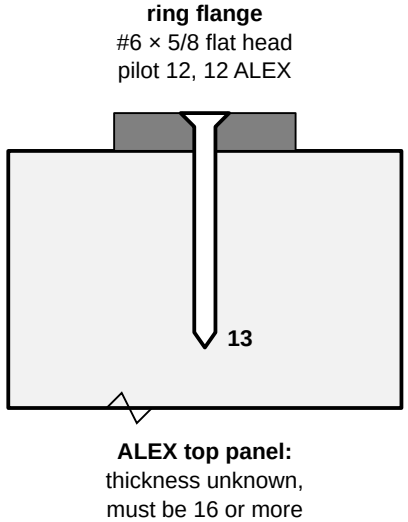


Fig. D-13 · Ring screws go into the ALEX top, not the desk. **MEASURE ON HARDWARE** panel thickness.

Setting a puck

11 Loosen the chuck. Drop the bit tip-first into the SET tower's hole for that bit (45 for 1/8 in, 35 for 5/64 in), rest the chuck nose on the tower, tighten. Re-set every time you change bits.

12 Stand the puck flat on the surface (on the rail's flat leg for 25 RAIL), bit down its bore, drill until the chuck nose lands on the puck. The hole is then the depth written on the puck.

If your chuck nose is domed, drill one test hole in scrap and measure it first.